SURVEYING & LEVELLING · NOTE SL07

Plane Table Surveying — Equipment, Setting Up and Orientation

Course topics: SWCE1.4 L8

What plane tabling is

Plane tabling is a graphical method: plot the points directly on the map in the field, so field work and plotting happen together.

- Errors do not accumulate as in traversing.
- The surveyor sees the ground while drawing it.
- It suits small- and medium-scale mapping; do not expect high precision.

Equipment

Drawing board

- Seasoned teak or pine, 40 × 30 cm to 75 × 60 cm, with a smooth flat top, mounted on a tripod.
- You can level it, rotate it about a vertical axis and clamp it in any position.
- Level it by shifting the tripod legs; some tripods have levelling screws or a ball-and-socket head.

Alidade — for sighting and drawing rays

Type	Description
Peep-sight alidade	Most common in agricultural surveying. A straight-edge ruler 45–60 cm long with graduated bevelled edges and two vertical sight vanes — one with a narrow slit, the other with a vertical wire or hair
Telescopic alidade	A telescope (like a theodolite telescope) mounted on the straight edge, with a vertical circle, level tube, cross hairs and stadia hairs

Accessories

Accessory	Use
Trough compass	Marks the magnetic meridian
Spirit level	Levels the table
Forked plumb bob	Centres the table over the station
Waterproof cover	Protects the sheet from rain

Setting up the table

1. **Set up:** place the table about waist-high over the station with the tripod legs spread wide and firm.
2. **Centre:** bring the point on the paper vertically above the ground point with the forked plumb bob (or drop a stone from the underside of the board). The accuracy needed depends on the map scale — for scales smaller than 1 cm = 6 m, rough centring is enough.
3. **Level:** place the spirit level first parallel to one side, then perpendicular to it.

4. **Orient:** turn the board until every line on the paper is parallel to the matching line on the ground, then clamp it.

Methods of orientation

Method	How	When
By compass	Place the trough compass along the meridian drawn earlier; rotate the board until the needle reads zero; clamp	Small-scale, rough mapping
By back sighting	Place the alidade along a previously plotted line; rotate the board until the line of sight falls on the previous station; clamp	More accurate — preferred

Advantages and disadvantages

Advantages	Disadvantages
No field notes needed; the fastest method of surveying	Not suitable in a wet climate
Errors in measuring and plotting are spotted at once; the plan is compared with the ground on the spot	Equipment is heavy and awkward to carry
Details cannot be overlooked; works in magnetic areas (where a compass fails); needs little skill	Not suitable for very accurate work

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The most accurate method of orienting a plane table is

(a) by compass
(b) by back sighting
(c) by resection only
(d) by eye

Q2. The alidade most used in agricultural surveying is the

(a) telescopic alidade
(b) peep-sight alidade
(c) micrometer alidade
(d) digital alidade

Q3. A forked plumb bob is used for

(a) levelling the table
(b) centring the table
(c) orienting the table
(d) drawing rays

Q4. Plane tabling is particularly suitable where

(a) the climate is wet
(b) high precision is needed
(c) magnetic disturbance makes a compass unreliable
(d) areas are very large

Answers

Q1 — (b) by back sighting
Q2 — (b) peep-sight alidade
Q3 — (b) centring the table
Q4 — (c) magnetic disturbance makes a compass unreliable